

Sriram Sattiraju

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EDUCATION

University of Texas at Austin

Austin, TX

Bachelor of Science in Computer Science

Expected Graduation: May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Computer Architecture, Operating Systems, Programming Languages, Cloud Computing, Discrete Math, Linear Algebra, Number Theory.

EXPERIENCE

Software Engineering Intern

June 2026 – Aug. 2026

Soularis

Austin, TX

- Built a pgvector retrieval engine (HNSW-indexed), threshold-tuned to reach **NDCG@5 70%** & **Recall@5 74%**.
- Launched RAG pipeline with 4o-mini + grounding validation, achieving **100%** schema compliance at **2.3s** latency.
- Unified 25 API endpoints into a single Railway schema, reducing **parsing logic by 20%** & **error rate by 15%**.

Software Engineering Intern

May 2025 – Sep. 2025

KindleTech Labs

London, UK (Remote)

- Deployed **16 FastAPI endpoints** for **PyTorch** GAN telecom anomaly data generation, achieving **~90%** fidelity.
- Optimized Flower federated pipeline latency on **AWS EC2** to **200ms** via in-memory loading + response caching.
- Built **Ollama** LLM query interface via **FastAPI** for real-time telecom diagnostics, deployed to **10+** clients.

Undergraduate Research Assistant

Sep. 2023 – June 2025

University of North Texas

Denton, TX

- Hosted **Flask** backend on EC2 with **Deepgram** + **OpenAI Whisper**, achieving **~1.5s** speech-to-text latency.
- Architected a **Unity** Mirror networking server supporting **20+** concurrent sessions with ID-based scene routing.
- Built real-time EEG streaming pipeline via **Mosquitto MQTT** + WebSockets, achieving **512ms** signal latency.

PROJECTS

BevoFit | React Native, Supabase, AWS Lambda, BeautifulSoup4, Prisma ORM

Oct. 2025 – Present

- Launched fitness platform with **500+ users** and **1,000+ daily API requests** on Supabase & React Native.
- Automated 5 RecSports scrapers with BeautifulSoup4 via **AWS Lambda** + EventBridge, at **240+ days** uptime.
- Implemented local SQLite cache via **Prisma**, syncing RecSports data on launch to minimize redundant API calls.

Source | TypeScript, VSCode Extension API, MiniSearch, Cheerio, MCP

Mar. 2026

- Built VSCode extension indexing docs and injecting ranked context into LLM agents, reducing hallucination **40%**.
- Engineered sitemap-first crawler with DOM fallback, indexing **400+** URLs into structured markdown in **~20s**.
- Built MCP server with **BM25** + **MiniSearch** retrieval, returning top-5 relevant docs to agent context in **~40ms**.

CUDA Tensor Engine | CUDA C++, C++

June – Aug. 2026

- Built GPU-accelerated tensor library in **CUDA C++** with custom memory-management and **RAII** ownership.
- Implemented CUDA kernels for matmul & reduction ops, achieving a **69.44x** speedup over CPU on 1024x1024.
- Engineered graph-based reverse-mode autograd with topo sort, posting **3.56ms** forward + backward on 512x512.

TECHNICAL SKILLS

Programming: Python, C++, JavaScript, HTML, CSS, Java, C, C#, PL/SQL, PostgreSQL, Firebase, Supabase

Frameworks & Libraries: React, React Native, Tailwind, Matplotlib, NumPy, Node.js, Express, PyTorch, Pandas

Tools & Platforms: Postman, Git, VSCode, Docker, AWS, Kubernetes, Redis, Grafana, Figma, Google Analytics

Concepts: Object-Oriented Programming, REST APIs, CRUD, Model-View Controller, Object-Relational Mapping

Testing & Debugging: Pytest, Unit Testing, Test-Driven Development, Debuggers (gdb, Valgrind)

AI-Native & LLM: Langchain, LangGraph, agent design (tools, planning, memory), RAG, embeddings & vector search

Publications:

- “Cognitive Energy Modeling for Neuroadaptive Human-Machine Systems using EEG and WGAN-GP”
- “A Comprehensive Comparison Between ANNs and KANs for Classifying EEG Alzheimer’s Data”